

EKI-5524SS/MM Series

4-port + 2x100FX port (Single/Multi-mode, SC/ST type), Fast Ethernet ProView Switch



Features

- Communicates with SCADA software via Modbus/TCP
- Communicates with NMS (Networking management system) via SNMP
- Port-based QoS for deterministic data transmission
- 40~75°C operating temperature range (EKI-5524SSI/SSI-ST, EKI-5524MMI/MMI-ST only)
- 12~48 V_{DC} (8.4~52.8 V_{DC}) wide-range power input
- EMS level 3 protection for extreme outdoor environments
- IEEE 802.3az Energy Efficient Ethernet (EEE)
- Jumbo Frame Support (Up to 9,216 Bytes)
- Supports redundant 12~48V_{DC} power input and P-Fail relay
- Loop detection

Introduction

The EKI-5524SS/SSI/SS-ST/SSI-ST and EKI-5524MM/MMI/MM-ST/MMI-ST are the world's first convergence switches for process control and IT networking management. This series uses Modbus/TCP to communicate with the SCADA software and SNMP to communicate with NMS (Networking Management System) at the same time, thereby allowing full read control over the devices either for control engineers or for IT. The devices come with the Port-based QoS for deterministic data transmission allows the priority ports to prioritize the traffic coming over those ports and delay the less immediately necessary data over the remaining ports. EKI-5524SS/SSI/SS-ST/SSI-ST and EKI-5524MM/MMI/MM-ST/MMI-ST switches use the highest quality components, to enable the range to operate in temperatures of between -40 and 75°C along with EMS Level 3 protection to repel electromagnetic interface for industrial resistance.

Specifications

Communications

- Standard** IEEE 802.3, 802.3u, 802.3x, 802.1p, 802.3az, 802.3ab
- LAN** 10/100Base-T(X), 100Base-FX
- Transmission Distance** Ethernet: Up to 100 m
Multi-mode Fiber: Up to 2 km (EKI-5524MM Series)
Single-mode Fiber: Up to 30 km (EKI-5524SS Series)
- Optical Fiber**
 - Multi-Mode (EKI-5524MM/MMI/MM-ST/MMI-ST)**
Wavelength: 1310nm
Tx Power: -14/-20 dBm
Rx Sensitivity: -32 dBm
Parameters: 50/125 μ m, 62.5/125 μ m
 - Single-Mode (EKI-5524SS/SSI/SS-ST/SSI-ST)**
Wavelength: 1310 nm
Tx Power: -8/-15 dBm
Rx Sensitivity: -34 dBm
Parameters: 9/125 μ m
- Transmission Speed** Up to 100 Mbps

Interface

- Connectors** 4 x RJ45 ports
2 x SC/ST type fiber optic connectors
6-pin screw Terminal Block (including relay)
- LED Indicators** P1, P2, P-Fail, Loop detection
10/100T(X): Link/Activity, Speed

Switch Properties

- MAC Table Size** 2K
- Packet Buffer Size** 1M bit
- Switching Capacity** 1.2 Gbps
- Jumbo Frame** 9216 bytes

Power

- Power Consumption** Max. 4 W
- Power Input** 12~48 V_{DC} (8.4~52.8 V_{DC}), redundant dual inputs
- Fault Output** 1 Relay Output

Mechanism

- Dimensions (W x H x D)** 43 x 120 x 84 mm
- Enclosure** IP30, metal shell with solid mounting kits
- Mounting** DIN-Rail, Wall

Protection

- Reverse Polarity** Present
- Overload Current** Present

Environment

- Operating Temperature** EKI-5524SS/SS-ST/MM/MM-ST: -10~60°C (14~140°F)
EKI-5524SSI/SSI-ST/MMI/MMI-ST: -40~75°C (-40~167°F)
- Storage Temperature** -40 ~ 75°C (-40 ~ 167°F)
- Operating Humidity** 10 ~ 95% (non-condensing)
- Storage Humidity** 10 ~ 95% (non-condensing)
- MTBF** 144,890 hours

Certification

- Safety** IEC/EN60950, UL60950, UL508, Class 1 Division 2, ATEX
- EMI** FCC Part 15 Subpart B Class A, EN 55011/55022 Class A
- EMS** EN61000-4-2 (ESD) Level 3
EN61000-4-3 (RS) Level 3
EN61000-4-4 (EFT) Level 3
EN61000-4-5 (Surge) Level 3
EN61000-4-6 (CS) Level 3
EN61000-4-8 (Magnetic Field) Level 3
- Shock** IEC 60068-2-27
- Freefall** IEC 60068-2-32
- Vibration** IEC 60068-2-6

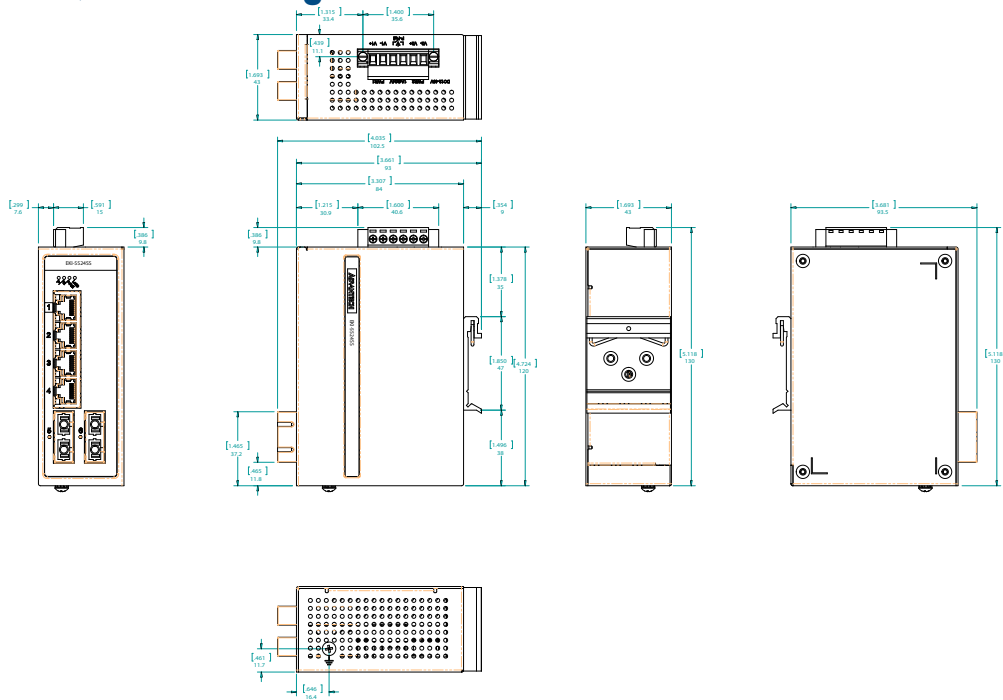
Ordering Information

- EKI-5524SS** 4-port +2x100FX port (Single -mode, SC type), Fast Ethernet ProView Switch
- EKI-5524SSI** 4-port + 2x100FX port (Single-mode, SC type), Fast Ethernet ProView Switch with Wide Temperature
- EKI-5524MM** 4-port + 2 x100FX port (Multi-mode, SC type), Fast Ethernet ProView Switch
- EKI-5524MMI** 4-port + 2 x100FX port (Multi-mode, SC type), Fast Ethernet ProView Switch with Wide Temperature
- EKI-5524SS-ST** 4-port + 2 x100FX port (Single-mode, ST type), Fast Ethernet ProView Switch
- EKI-5524SSI-ST** 4-port + 2 x100FX port (Single-mode, ST type), Fast Ethernet ProView Switch with Wide Temperature
- EKI-5524MM-ST** 4-port + 2 x100FX port (Multi-mode, ST type), Fast Ethernet ProView Switch
- EKI-5524MMI-ST** 4-port + 2 x100FX port (Multi-mode, ST type), Fast Ethernet ProView Switch with Wide Temperature

EKI-5524SS/MM Series

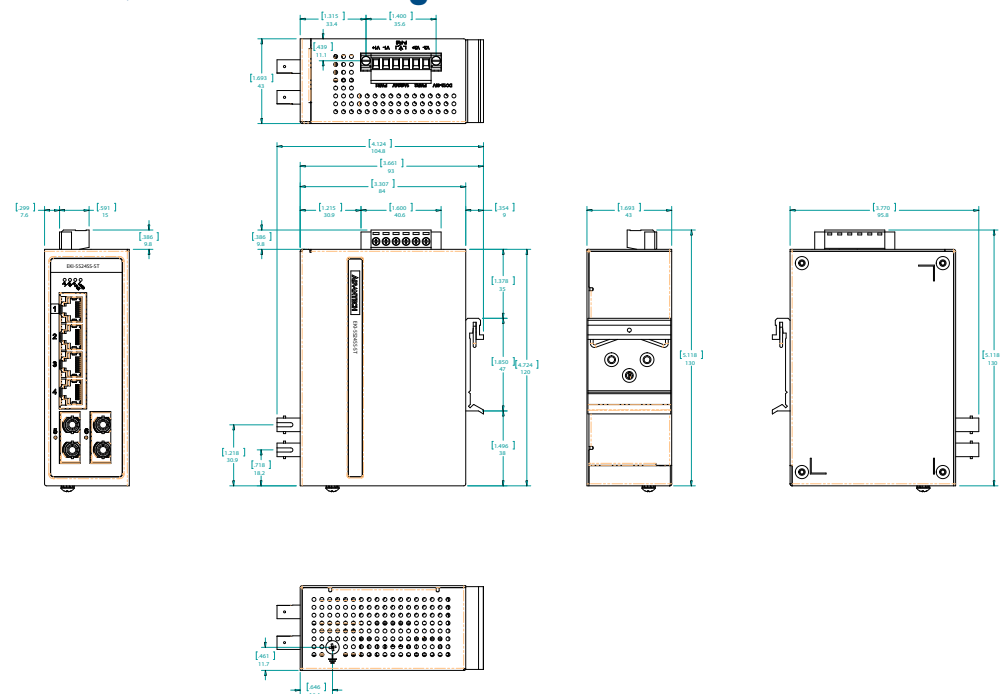
EKI-5524SS/SSI Mounting Dimensions

Unit: mm [inch]



EKI-5524SS-ST/SSI-ST Mounting Dimensions

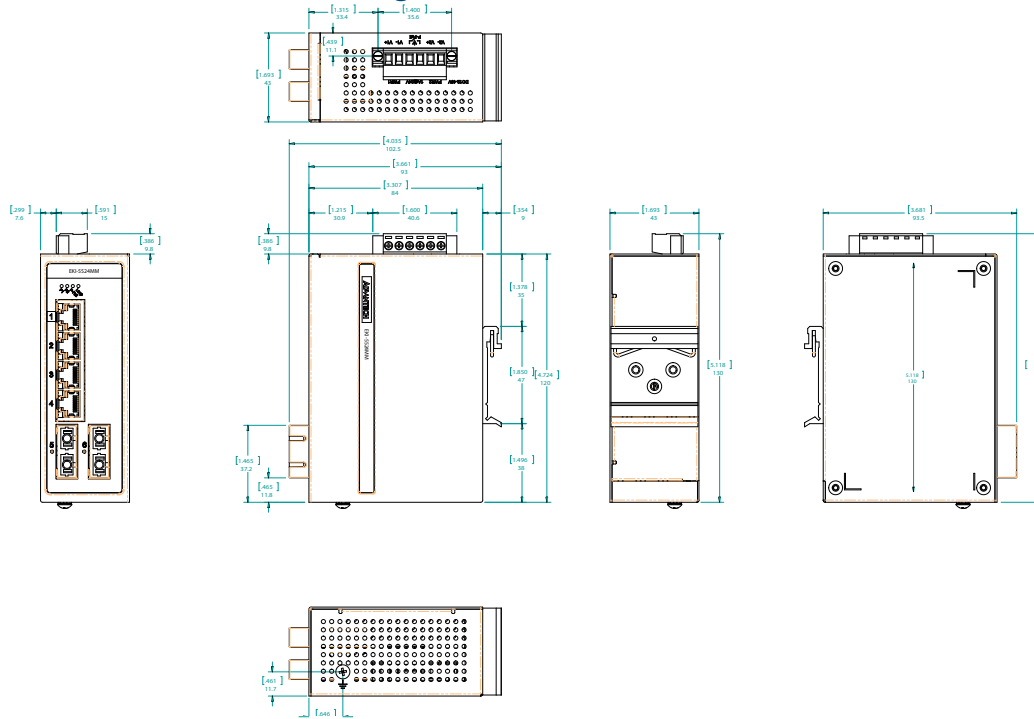
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EKI-5524SS/MM Series

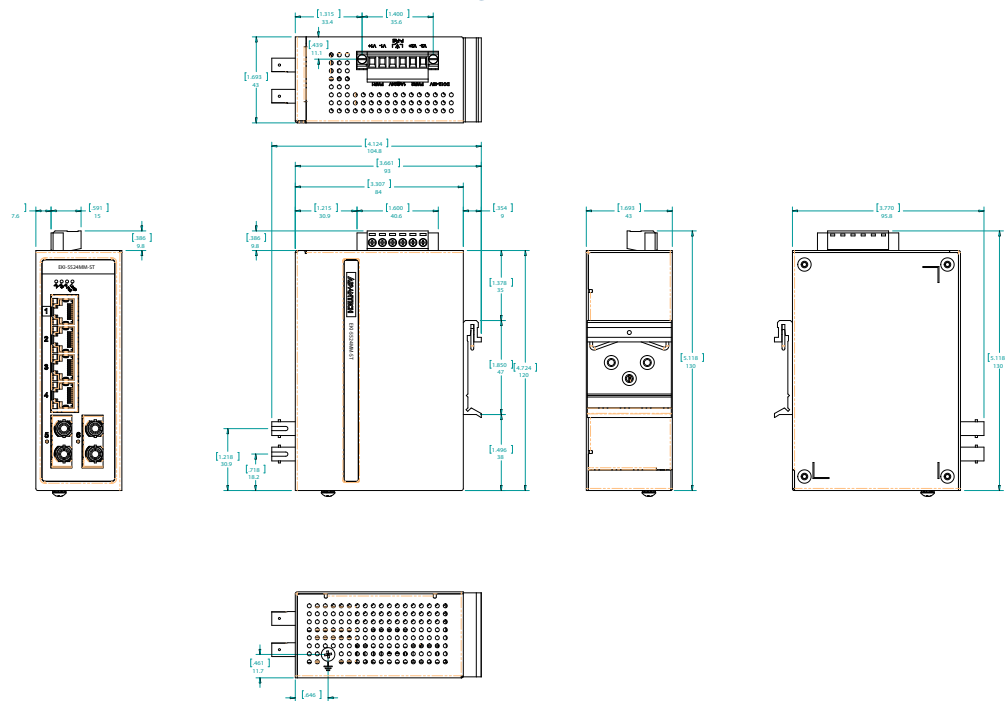
EKI-5524MM/MMI Mounting Dimensions

Unit: mm [inch]



EKI-5524MM-ST/MMI-ST Mounting Dimensions

Unit: mm [inch]



ECU-1051

Cloud enabled Intelligent Communication Gateway for IIOT application

NEW



Features

- TI Cortex A8 processor
- 2 x RS-232/485 serial ports
- 2 x 10/100 Ethernet ports
- 1 x Mini-PCle for WIFI/Cellular/4G
- Support web service for remotely on-line monitoring
- Supports SD cards and OTA firmware updates
- EdgeLink IoT software for data acquisition and cloud communication
- Wide Operating temperature: -40 ~ 70°C



EdgeHub



Introduction

For facility monitoring, distributed solar power, smart factory sector, especially cloud based applications which require a total wireless and Ethernet communication solutions, Advantech has released the ECU-1051 RISC-based industrial communication gateway. ECU-1051 has an open platform design with Cortex A8 processor, two 10/100 Ethernet ports and operating temperature range of -40~70°C. With Linux operating system and EdgeLink IoT software, ECU-1051 enable system integrators to develop applications which precisely fit their need.

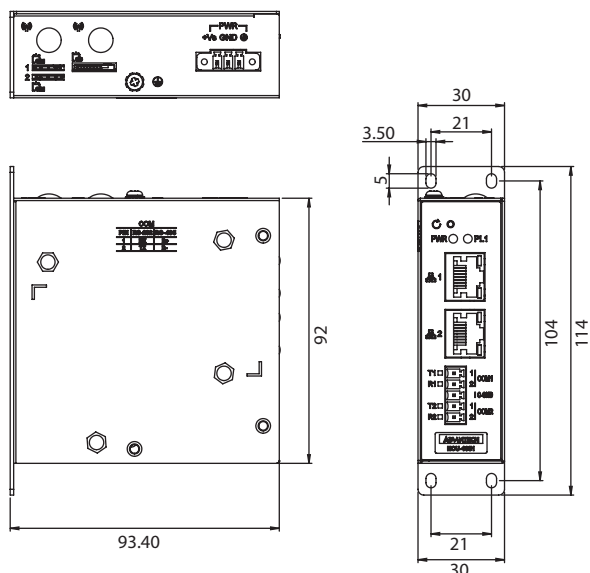
Specifications

System	CPU	ECU-1051TL-R10AAE		ECU-1051TL-R10AB	ECU-1051TL-R10ABU	ECU-1051E-R10AAE
	RAM			TI Cortex A8, 600MHz		TI Cortex A8, 300MHz
	Storage			DDR3L 256MB		DDR3L 128MB
	SD Slot			512MB NAND		
				1 x Micro-SD		
	SIM Slot			2 x Nano SIM		1 x Nano SIM
	LED Indicators			Power, LAN (LINK, ACT), Programmable		
	Real Time Clock			Yes		
Watchdog Timer			Yes			
TPM			NA			
I/O Interfaces	Serial Ports	2 x RS-232/485 (isolation optional), 5-Pin Terminal Block, 50 – 115.2kbps			2 x RS-232/485 (isolation 3000Voc) 5-Pin Terminal Block, 50 – 115.2kbps	2 x RS-232/485 (isolation optional) 5-Pin Terminal Block, 50 – 115.2kbps
	LAN Ports	2 x 10/100 Base-T Ethernet (RJ45)				
	USB Ports	NA	1 x USB 2.0			NA
	Displays	NA				
	Mini-PCIE	1x Mini-PCle (Full-size) for Wireless Module Signal: USB 2.0				
EdgeLink	Protocol Support	SouthBound			NorthBound	
		IEC 61850 MMS Client*			IEC 61850 MMS Server*	
		IEC 61850 GOOSE Client*			IEC 61850 GOOSE Server*	
		IEC 60870-5-101			Modbus Server	
		IEC 60870-5-103			IEC 60870-104 Server	
		IEC 60870-5-104			WASCADA (WebAccess)	
		DL/T 645-2007			BACnet Server	
		DL/T 645-1997			OPC UA Server	
		IEC 62056-21			Data Transfer	
		ODBC				
		JDBC				
		OPCUA Client				
	BACnet IP					
	BACnet MS/TP					
	SNMP					
	Modbus Client					
	PLC driver support	200+ PLC drivers ABB/Advantech/Allen-Bradley/BECKHOFF/ DELTA/FATEK/GE/Honeywell/Keyence/Mitsubishi/Omron/Panasonic/Schneider/Sharp/Siemens/Toyopuc/Wago 750/Yaskawa/Yokogawa				
	Programming Support	IEC 61131-3, Linux C, Python				
Configuration Tool	EdgeLink Studio					
Data Monitoring	2000 Tags MAX.				500 Tags MAX.	
Cloud Connectivity(MQTT)	EdgeHub, IoTSuite, Azure, AWS, Google Cloud IoT Core and more					
Database Transmission	SQL Server, MySQL, ORACLE, FTP Server					
VPN	Open VPN					
General	Certification	CE, FCC, UKCA				
	Dimensions (W x D x H)	93.7 x 92 x 30mm				
	Form Factor	Small Size				
	Enclosure	SGCC				
	Mounting	Wall-Mount/DIN-Rail				
	Weight(Net)	0.6kg				
	Power Requirement	10 – 30 Vdc				
	Power Consumption	1.2W @ 24Vdc				
	Power Connector	1 x 3-Pin Terminal Block				
Environment	Operating Temperature	-40~70°C				
	Storage Temperature	-40~85°C				
	Relative Humidity	95% (non-condensing)				
	Ingress Protection	IP30				
	Vibration Protection	2Grms, Random/Sine, 5 – 500Hz				

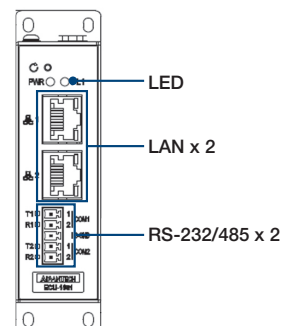
* Please refer to the EdgeLink specification for further details. Activation of the IEC-61850 protocol requires an additional fee.

Dimensions

Unit: mm

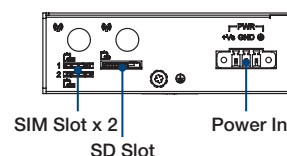


Front View

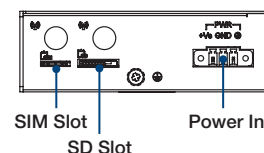


Top View

ECU-1051



ECU-1051E



Ordering Information

Part Number	CPU	RAM	NAND	OS	IoT Software	TPM 2.0	LAN	COM	USB	Console	Country of Origin
ECU-1051TL-R10AAE	TI Cortex A8, 600MHz	DDR3L 256MB	512MB	Yocto	EdgeLink	N/A	2	2	0	3pin COM on board	China
ECU-1051TL-R10AB									1		China
ECU-1051TL-R10ABU									1		Taiwan
ECU-1051E-R10AAE	TI Cortex A8, 300MHz	DDR3L 128MB							0		China

Optional Accessory

Cellular Module and Antenna

- 96PD-EC25EFA LTE Module Cat. 4, Mini-PCle for EU,TW,IN,KR,SEA
- 96PD-EC25JFA LTE Module Cat. 4, Mini-PCle for JP
- 96PD-EG25GGB LTE Module Cat. 4, Mini-PCle for Global
- 1750006264 SMA(F) Cable, 15 cm
- 1750005865 Dipole Antenna, 11 cm
- 96PD-MDG200AU0 AMIC 4G.Cat4 Socket Modem(US)
- 1750006264 SMA(F) Cable, 15 cm
- 1750008303-01 Antenna, 13cm

Wifi Module and Antenna

WIFI

- 96PD-RYUW131A1 Wi-Fi 4 2.4GHz mPCIe with Antenna and cable

WIFI+BT:

- EWM-W172H01E 802.11ac/a/b/g/n 2T2R Wi-fi + Bluetooth 4.2 Half
- 1750008717-01 Wi-Fi 2.4GHz+5GHz dipole antenna
- 1750006043 SMA(M) Coaxial Cable, 150 mm
- 9656EWMG00E Half-size miniPCIe to miniPCIe bracket set

Micro SD Card

- 96FMMSDI-4G-ET-AT1 ATP 4G MICRO SD CARD I-GRD SLC(G)
- 96FMMSDI-8G-ET-AT1 ATP 8G MICRO SD CARD I-GRD SLC(G)

DIN-Rail Power Supply

- 96PSD-A20W24-MM A/D 100-240V 20W 24V Power Supply
- PSD-A40W24 DIN RAIL A/D 100-240V 40W 24V Power Supply

Adapters and Power Cables

- 96PSA-A60W24T2-3 AC100-240V to DC24V, 60W Adapter, C14 CORD END 2PIN
- 1702002600 Power cable (US) 1.8 M
- 1702002605 Power cable (EU) 1.8 M
- 1702031801 Power cable (UK) 1.8 M
- 1700000596 Power cable (China/Australia) 1.8 M